

INTERNSHIP / TRAINING REPORT

ON

**CISCO NETWORKING ACADEMY TRAINING:
NETWORKING, CYBERSECURITY, IoT AND DIGITAL
TRANSFORMATION**

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



Submitted By
ANKUSH RAJ
Roll Number: 2410031647
5th Semester, 3CSE9

Academic Year: 2026-27

CANDIDATE'S DECLARATION

I, ANKUSH RAJ, hereby declare that the training report titled "Cisco Networking Academy Training: Networking, Cybersecurity, IoT and Digital Transformation" has been completed by me to fulfil the academic requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report documents my learning and training experience through three courses completed on Cisco Networking Academy: Networking Basics, Introduction to Cybersecurity, and Introduction to IoT and Digital Transformation.

The technical descriptions and course information included in this report have been prepared from the course materials and official Cisco Networking Academy references cited in the bibliography. The certificates included in this report are my course-completion certificates. I have not represented the training as employment or professional experience at Cisco.

I affirm that this report is my own academic work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation arising from material submitted by me.

Signature: _____

Student Name: ANKUSH RAJ

Roll No.: 2410031647

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me during the completion of my Cisco Networking Academy training and the preparation of this report. The courses provided a structured way to strengthen my understanding of computer networking, cybersecurity, Internet of Things and digital transformation.

I am thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing an academic environment in which I could pursue additional technical learning alongside my B.Tech CSE curriculum.

I also acknowledge Cisco Networking Academy for providing accessible, structured learning resources, interactive activities, assessments and course-completion credentials. The learning experience helped me connect theoretical concepts with practical technology applications and identify areas for further study.

Finally, I express my gratitude to my parents and everyone who encouraged me throughout this learning journey.

Signature: _____

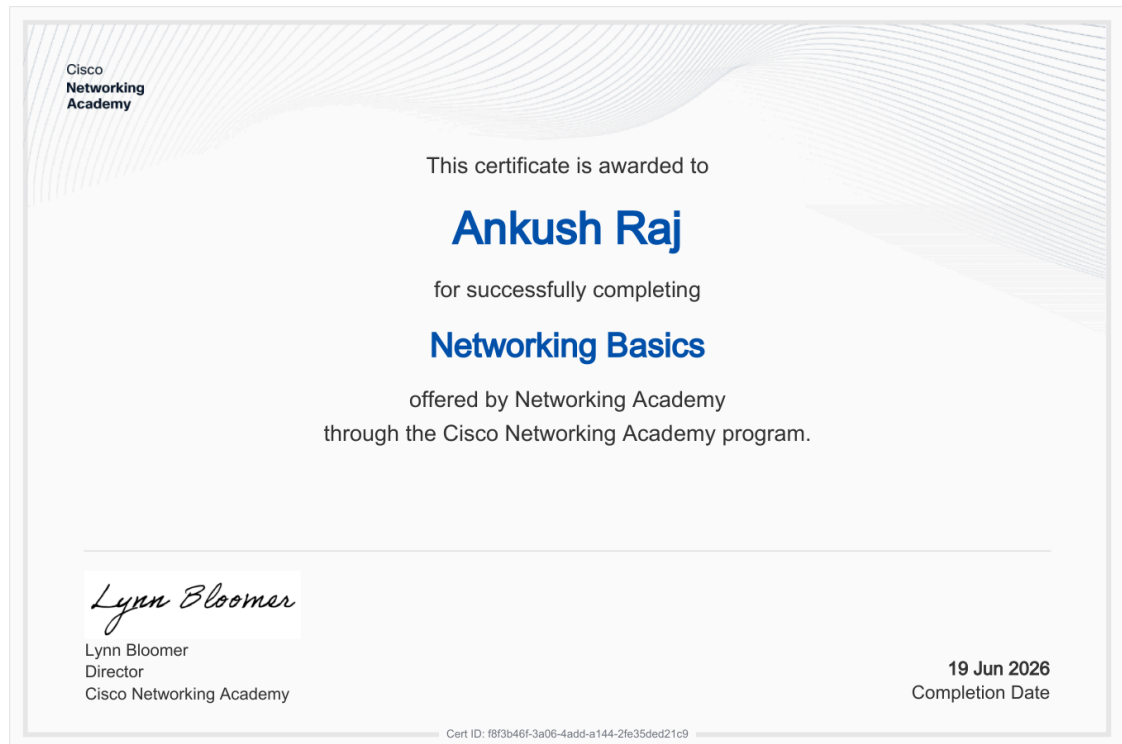
Student Name: ANKUSH RAJ

Roll No.: 2410031647

CERTIFICATES OF COMPLETION

The following three certificates document successful completion of Cisco Networking Academy courses. The original certificates are reproduced below as evidence of course completion. Certificate details were read directly from the uploaded certificates.

Networking Basics



Introduction to Cybersecurity

Cisco
Networking
Academy

This certificate is awarded to

Ankush Raj

for successfully completing

Introduction to Cybersecurity

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

Lynn Bloomer
Director
Cisco Networking Academy

18 Jun 2026
Completion Date

Cert ID: ba50ba94-b82e-46ac-a506-fe2596b127fb

Introduction to IoT and Digital Transformation

Cisco
Networking
Academy

This certificate is awarded to

Ankush Raj

for successfully completing

Introduction to IoT and Digital Transformation

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

Lynn Bloomer
Director
Cisco Networking Academy

18 Jun 2026
Completion Date

Cert ID: c62c6489-bdfc-4ec1-93e5-333ff65beca2

TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate	-
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	2
4.3 Problem Statement	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used.....	4
4.7 System Architecture	5
4.8 Methodology	6
4.9 Expected Outcomes.....	7
4.10 Certificates of Completion and Communication Proof	8
5. Bibliography/References	9

CHAPTER 4 – PROJECT / TRAINING DESCRIPTION

4.1 Introduction

The training documented in this report was completed through Cisco Networking Academy and focused on three complementary introductory technology courses: Networking Basics, Introduction to Cybersecurity, and Introduction to IoT and Digital Transformation. The three courses were completed in June 2026. The certificates show completion of Networking Basics on 19 June 2026 and completion of the other two courses on 18 June 2026.

The purpose of combining these courses was to develop a broad foundation rather than studying each area in isolation. Networking provides the communication foundation, IoT demonstrates how physical and digital devices become connected, and cybersecurity explains the need to protect information, devices and digital systems. This progression provides a useful starting point for further study in network administration, cybersecurity, cloud computing, IoT and software development.

4.2 Organization Profile

Cisco Networking Academy is an education and skills-development program associated with Cisco that provides technology learning opportunities in areas such as networking, cybersecurity and emerging technologies. The official Cisco course material describes Networking Academy courses as structured learning experiences that can be delivered through instructor-guided or self-paced formats, depending on the course.

For this training, Cisco Networking Academy served as the learning platform and course provider. The three selected courses were introductory in nature and were suitable for building foundational technical awareness. Cisco's published course information for Introduction to Cybersecurity identifies topics including cyber trends, threats, staying safe in cyberspace and protecting personal and company data. The IoT course introduces the Internet of Things and the effect of digital transformation on organizations, industries and daily life.

4.3 Problem Statement

Modern computing systems depend on interconnected networks, connected devices and digital services. A student entering the field of computer science therefore needs an understanding of how devices communicate, how digital information can be

exposed to threats, and how emerging technologies such as IoT, artificial intelligence and data analytics are changing technology environments.

The training addressed this learning need by providing structured introductory coursework across networking, cybersecurity and IoT/digital transformation. The objective was not to claim professional-level expertise, but to establish a reliable conceptual foundation and identify areas for continued technical development.

4.4 Project / Training Objectives

- To understand the fundamental concepts of computer networking and network communication.
- To become familiar with network devices, media, protocols and basic addressing concepts.
- To develop awareness of cybersecurity threats, online safety and data protection.
- To understand the basic role of cybersecurity in connected digital environments.
- To explore the Internet of Things and the concept of connected devices.
- To understand how data, artificial intelligence and other emerging technologies contribute to digital transformation.
- To connect networking, IoT and cybersecurity concepts into a broader view of modern IT systems.
- To document the completed learning and certificates in a structured academic report.

4.5 Scope of the Training

The scope of this training is foundational and educational. It covers the concepts introduced in the three completed courses and the knowledge gained from their learning activities and assessments. It does not claim advanced network engineering, professional penetration testing, production IoT deployment, or industry-level security operations.

The knowledge gained can serve as a base for future study in networking, cybersecurity, cloud computing, IoT, software development and related areas. The training is particularly useful for understanding how different technology domains interact in a modern digital environment.

4.6 Technologies and Tools Used

Technology / Tool	Role in Learning
Cisco Networking Academy	Primary online learning platform and course delivery environment.
Computer Networking Concepts	Understanding network communication, devices, media and protocols.
IP Addressing Concepts	Understanding how devices are identified and communicate on IP networks.
Cybersecurity Concepts	Understanding threats, online safety, data and privacy protection.
Internet of Things (IoT)	Understanding connected devices, data and digital transformation.
Data Analytics	Understanding the role of data in emerging technology and digital transformation.
Artificial Intelligence	Understanding AI as an emerging technology connected with IoT and digital transformation.
Cisco Packet Tracer activities	Where included by the IoT course structure, used as a network simulation activity.
Quizzes / Knowledge Checks	Used to assess understanding during the courses.
Final Assessments	Used to evaluate course-level learning and completion.

4.7 System / Learning Architecture

Because this work is a training-based project rather than a software product, the architecture is represented as a learning architecture. The learning flow connects the learner, Cisco Networking Academy content, practical activities and assessments to the final learning outcomes.

Stage	Description
1. Learner	Student accesses the assigned course content through Cisco Networking Academy.
2. Course Modules	Concepts are studied through structured modules, explanations and learning resources.
3. Activities	Interactive activities, practice items, knowledge checks and course-specific labs support learning.
4. Assessment	Quizzes and final assessments are used to evaluate understanding.
5. Completion	Successful completion results in the corresponding course-completion credential.
6. Learning Outcome	Student gains foundational knowledge that can support further technical study.

4.8 Methodology

The training followed a structured self-learning methodology. First, the course modules were studied in the sequence provided by Cisco Networking Academy. Important terminology and concepts were reviewed during learning. Where activities or simulations were available, they were used to connect theoretical ideas with practical examples. Knowledge checks and assessments were then used to evaluate understanding.

The three courses were approached as complementary learning units. Networking Basics was used to establish the communication foundation. Introduction to

Cybersecurity was then used to understand security concerns in digital systems. Introduction to IoT and Digital Transformation broadened the learning toward connected devices, data and emerging technologies.

Finally, the certificates were reviewed and organized as evidence of completion, and the learning outcomes were documented in this academic report.

4.9 Learning Outcomes

After completing the three courses, the following foundational outcomes were achieved:

- Understanding of basic computer networking terminology and communication concepts.
- Awareness of network devices, media and protocols used to enable communication.
- Improved understanding of IP-based communication and the role of addressing.
- Awareness of common cybersecurity concerns and the importance of protecting information.
- Understanding of safe digital behavior and the need to protect personal and organizational data.
- Introductory understanding of IoT, connected devices and data-driven systems.
- Awareness of how artificial intelligence and data analytics can contribute to digital transformation.
- Ability to relate networking and cybersecurity concepts to connected IoT environments.
- Improved understanding of possible future learning and career directions in networking, cybersecurity and emerging technologies.

4.10 Certificates of Completion and Other Evidence

The three certificates included in this report provide evidence of successful completion of the selected Cisco Networking Academy courses. The certificate details are summarized below.

Course	Completion Date	Certificate ID
Networking Basics	19 June 2026	f8f3b46f-3a06-4add-a144-2fe35ded21c9

Introduction to Cybersecurity	18 June 2026	ba50ba94-b82e-46ac-a506-fe2596b127fb
Introduction to IoT and Digital Transformation	18 June 2026	c62c6489-bdfc-4ec1-93e5-333ff65beca2

The original certificate copies are included in the Certificates section of this report and are also stored separately in the accompanying GitHub repository. The repository is organized so that the three certificates, this report and the presentation can be accessed independently.

The screenshot displays the Cisco Networking Academy Learning History page. The interface includes a search bar, filters for completion status and sort order, and a list of three courses. The first course, 'Networking Basics', is marked as 'IN PROGRESS' and completed on June 19, 2026. The second course, 'Introduction to IoT and Digital Transformation', is also marked as 'IN PROGRESS' and completed on June 18, 2026. The third course, 'Introduction to Cybersecurity', is marked as 'IN PROGRESS' and completed on June 18, 2026. The user's name 'Ankush Learner' is visible in the top right corner.

Summary of Individual Courses

Networking Basics

The Networking Basics course introduced the foundation of networking and network devices, media and protocols. The course helped me understand how data flows through a network, how devices communicate, and how networking applications and protocols support common networking tasks. The learning established a base for future study of network administration, troubleshooting and cybersecurity.

Introduction to Cybersecurity

The Introduction to Cybersecurity course explored cyber trends and threats, safe behavior in cyberspace and protection of personal and company data. The course helped me recognize that cybersecurity is a shared responsibility involving technical controls as well as awareness and responsible digital behavior.

Introduction to IoT and Digital Transformation

The Introduction to IoT and Digital Transformation course provided an exploratory view of the Internet of Things and how digital transformation affects organizations,

businesses, governments, industries and everyday life. Cisco's course information highlights the relationship of IoT with emerging technologies such as data analytics, artificial intelligence and cybersecurity. The course helped me understand the broader technology ecosystem in which connected devices generate and exchange information.

Conclusion

The Cisco Networking Academy training was a valuable foundational learning experience. Completing Networking Basics, Introduction to Cybersecurity, and Introduction to IoT and Digital Transformation helped me develop a connected understanding of networking, security and emerging technologies.

The most important outcome was not only the completion of three certificates but also the ability to relate the three areas: networking enables communication, IoT extends connectivity to devices and systems, and cybersecurity helps protect the resulting digital environment. This foundation will support my future learning in more advanced networking, cybersecurity, cloud, IoT and software-development topics.

CHAPTER 5 – BIBLIOGRAPHY / REFERENCES

Cisco Networking Academy. Introduction to Cybersecurity. Cisco Public course information. Available at: <https://www.netacad.com/courses/introduction-to-cybersecurity>

Cisco Networking Academy. Introduction to IoT and Digital Transformation. Cisco Networking Academy course catalog and course page. Available at: <https://www.netacad.com/courses/introduction-iot>

Cisco Networking Academy. Networking Basics. Cisco Networking Academy course information. Available at: <https://www.netacad.com/courses/networking-basics>

Cisco Networking Academy. Course Catalog, January 2024. Cisco Public Information.

Cisco Networking Academy. Introduction to Cybersecurity – course overview and learning components. Cisco Public Information.

Cisco Networking Academy. Introduction to IoT and Digital Transformation – course overview and learning components. Cisco Public Information.

Appendix: Repository Organization

The accompanying GitHub repository is organized to keep the evidence and academic deliverables easy to access:

2024-28_Ankush_Raj_2410031647_5thSemester_3CSE9/

├── README.md

├── Certificates/

| ├── Networking_Basics.pdf

| ├── Introduction_to_Cybersecurity.pdf

| └── Introduction_to_IoT.pdf

├── Report/

| └── Internship_Report.pdf

└── PPT/

└── Internship_Presentation.pptx